

Requirements Gathering

Movie Search Application

Project	API Used	Section
Movie Search App	Movie Database API (To Be Selected)	Requirements Gathering

Stakeholder Analysis

A stakeholder is anyone who has an interest in the project — whether they use it, build it, or evaluate it. Below are the main stakeholders for this app and what each one needs.

Stakeholder	Who Are They?	What Do They Need?
End User	Anyone who wants to search for movies	Simple search, fast results, clear movie info
Development Team	The students building the app	Clear requirements and a working API key
Project Supervisor	The doctor who grades the project	Good documentation and all requirements covered
API Provider	The company providing the movie data	Correct API usage and proper attribution

User Stories & Use Cases

A user story describes a feature from the user's point of view using this format: "As a [user], I want to [do something], so that [reason]."

User Stories

US-01 – Search for a Movie

As a user, I want to type a movie name and see a list of results, so that I can find the movie I'm looking for quickly.

Acceptance Criteria:

- ✓ There is a search bar on the home page
- ✓ Results appear after clicking Search or pressing Enter
- ✓ Each result shows the movie poster, title, and year
- ✓ If nothing is found, a message says "No results found"

US-02 – View Movie Details

As a user, I want to click on a movie and see its full details, so that I can decide if I want to watch it.

Acceptance Criteria:

- ✓ Clicking a movie card opens a details page
- ✓ The page shows: title, poster, description, rating, genre, release date
- ✓ There is a back button to return to the results

US-03 – Handle Errors Gracefully

As a user, I want to see a helpful message when something goes wrong, so that I know what to do next.

Acceptance Criteria:

- ✓ If the search is empty, show a message asking the user to type something
- ✓ If the API fails, show a friendly error message (not a blank page)

Use Cases (Summary)

Use Case	Actor	What Happens
UC-01: Search Movies	User	User types a title → app calls Movie Database API → results are shown
UC-02: View Details	User	User clicks a movie → app fetches details → details page opens
UC-03: Empty Search	User	User submits empty search → app shows validation message
UC-04: API Error	System	API call fails → app shows a friendly error message to the user

Functional Requirements

Functional requirements describe what the system must do. These are the actual features the app needs to have.

Search Feature

ID	Requirement	Priority
FR-01	The app must have a search bar on the home page	High
FR-02	Search must work when the user presses Enter or clicks the Search button	High
FR-03	The app must call the Movie Database API to get search results	High
FR-04	Each result must show the movie poster, title, and release year	High
FR-05	If the search box is empty, the app must show a warning message	Medium
FR-06	If no movies are found, the app must show "No results found"	Medium

Movie Details Page

ID	Requirement	Priority
FR-07	Clicking a movie must open a details page for that movie	High
FR-08	The details page must show: title, poster, description, rating, genres, release date	High
FR-09	There must be a back button to go back to the search results	High
FR-10	Show cast information if it is available from the API	Medium

Other Requirements

ID	Requirement	Priority
FR-11	Show a loading spinner while waiting for the API response	Medium
FR-12	If there are many results, show pagination (Next / Previous) buttons	Medium
FR-13	The API key must be stored safely and not visible in the code	High

Non-Functional Requirements

Non-functional requirements describe how the system should behave — things like speed, security, and ease of use. They are about quality, not features.

Performance

- Search results should appear within 2 seconds under a normal internet connection.
- The movie details page should load in under 2 seconds.
- The app must not send more than 40 requests per 10 seconds to the Movie Database API.

Security

- The Movie Database API key must be stored in an environment variable — never written directly in the code.
- All API requests must use HTTPS.
- User input must be checked before being sent to the API to avoid bad requests.

Usability

- The app must work properly on mobile phones, tablets, and desktops.
- A first-time user should be able to search for a movie and view its details without any help.
- Error messages must be written in simple language — no technical terms shown to the user.

Reliability

- If the Movie Database API is down, the app must show a friendly error page instead of crashing.
- The app should work on Chrome, Firefox, and Edge browsers.
- The code should be clean and well-commented so it is easy to update later.

Summary

Section	What Was Covered
Stakeholder Analysis	Identified 4 stakeholders and their needs
User Stories & Use Cases	3 user stories and 4 use cases
Functional Requirements	13 functional requirements (FR-01 to FR-13)
Non-Functional Requirements	12 quality requirements covering performance, security, usability, and reliability